Grade 11 C10: Statistical Model of Student Heights

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11 – Example | Global Economics 2026–2027

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1 Choose a question and obtain data

Question. What proportion of surveyed students is taller than 180 cm, and how might the estimate inform preliminary school-uniform inventory planning? The 180 cm cutoff is illustrative, not an official uniform specification. Height alone does not determine clothing size.

Let X denote a student's recorded height in centimeters. It is treated as continuous, although reported measurements may be rounded.

Table 1: Dataset summary

Characteristic	Value
Data source	Manually entered survey dataset
Variable	Recorded height (cm)
Observations	$n = 109$ numerical values
Observed range	151–191 cm; sample extrema, not population bounds

Data reference. Manually entered survey dataset.

2 Summarise the data

For observations $x_1, \ldots, x_n$, the sample statistics use denominator $n - 1$:

$$\bar{x} = \frac{\sum x_i}{n}, \qquad s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}, \qquad s = \sqrt{s^2}.$$

Table 2: Observed totals and sample statistics

Quantity	Calculation	Result
n	All retained records	109
$\sum x_i$	Direct sum	18 482.5 cm
$\sum x_i^2$	Direct sum	3 142 981.25 cm^2
$\sum (x_i - \bar{x})^2$	$3\,142\,981.25 - (18\,482.5)^2/109$	9 010.550459 cm^2
$\bar{x}$	$18\,482.5/109$	169.564220 cm
s^2	$9\,010.550459/108$	83.431023 cm^2
s	$\sqrt{83.431023}$	9.134058 cm

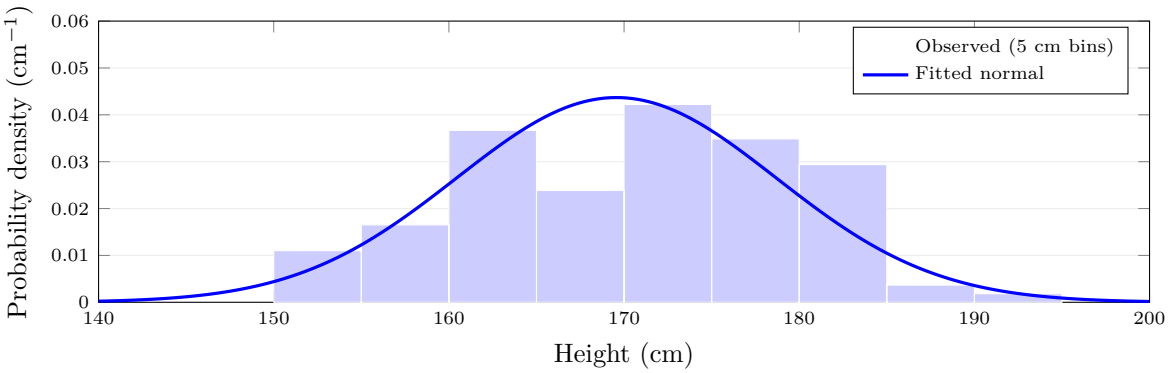


Figure 1: Sample-SD normal fit over the observed histogram. Bins are $[150, 155), \ldots, [190, 195)$; each bar height is its count divided by 109×5. Bars have total area one; curves and bars share density units. Both density overlays use identical axes and bins.

3 Select and fit one probability density function

The 5 cm histogram in Figure 1 is irregular and not perfectly symmetric. Following the assignment's specified normal-estimation rule, $\widehat{\mu} = \bar{x}$ and $\widehat{\sigma} = s > 0$.

$$X \sim N(169.564220\,\text{cm},\ 83.431023\,\text{cm}^2),$$
$$f(x) = \frac{1}{9.134058\sqrt{2\pi}} \exp\left[-\frac{(x - 169.564220)^2}{2(9.134058)^2}\right].$$

Here x is the numerical height in cm, $f(x)$ has units cm^{-1}, and the support is $(-\infty, \infty)$.

Table 3: Model parameters and moment comparison

Quantity	Sample	Fitted normal	Units
Mean	169.564220	169.564220	cm
Variance	83.431023	83.431023	cm^2
Standard deviation	9.134058	9.134058	cm

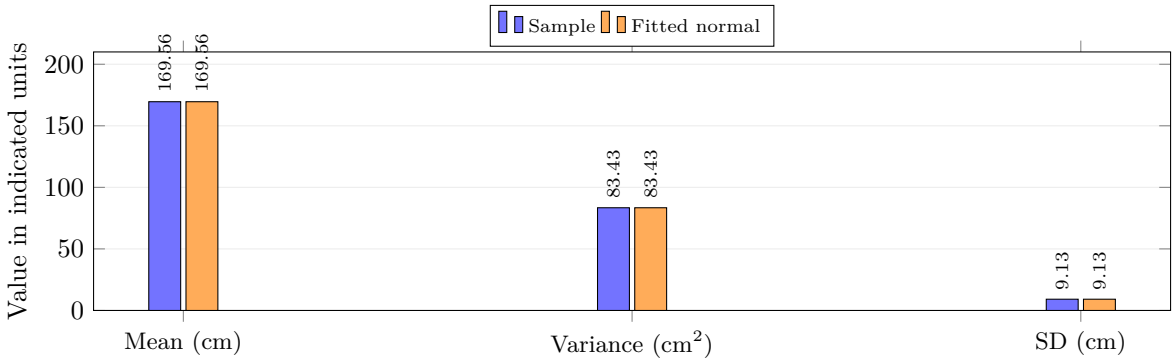


Figure 2: Paired sample–model moments: equal by parameter estimation, not independent evidence of fit. Bar labels are rounded to two decimal places; values with greater precision appear in Table 3. Each category uses the units indicated on the horizontal axis.

4 Calculate quantiles and model probabilities

Write Φ for the standard-normal CDF. With $z_{0.10} = -1.2815515655$ and $z_{0.90} = 1.2815515655$,

$$q_p = \widehat{\mu} + \widehat{\sigma} z_p,$$
$$q_{0.10} = 169.564220 + 9.134058(-1.2815515655) = 157.858453\,\text{cm},$$

$$q_{0.90} = 169.564220 + 9.134058(1.2815515655) = 181.269987\,\text{cm}.$$

The unrounded fitted quantiles imply model probabilities 0.10, 0.80, and 0.10 by construction. Endpoints have probability zero.

For empirical comparisons, the fixed cutoffs are $a = 157.858453\,\text{cm}$, $b = 181.269987\,\text{cm}$ and $c = 180.000000\,\text{cm}$. All model probabilities below are evaluated at *these same numerical cutoffs*, using the full-precision fitted moments.

$$P(X < c) = \Phi\left(\frac{c - \widehat{\mu}}{\widehat{\sigma}}\right),$$

$$P(X > 180) = 1 - \Phi\left(\frac{180 - 169.564220}{9.134058}\right) = 0.126620437.$$

5 Compare empirical and model probabilities

Table 4: Observed versus fitted probabilities at identical boundaries (cm)

Event (cm)	Count	n	Empirical	Model	Absolute	Error (%)
$X < 157.858453$	10	109	0.091743	0.100000	0.008257	8.26
$157.858453 \le X \le 181.269987$	91	109	0.834862	0.800000	0.034862	4.36
$X > 181.269987$	8	109	0.073394	0.100000	0.026606	26.61
$X > 180.000000$	13	109	0.119266	0.126620	0.007354	5.81

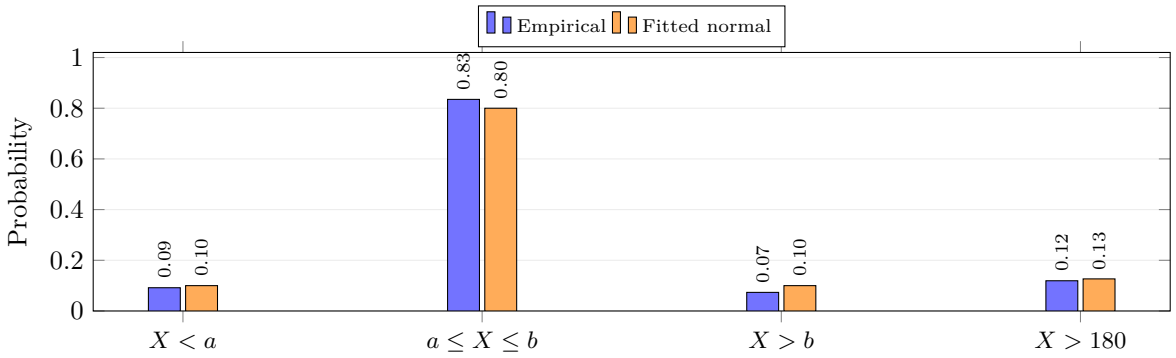


Figure 3: Paired empirical and model probabilities, with matching event definitions. Values are rounded to two decimals in the bar labels; Table 4 retains six decimals.

Counts are based on individual observations, not histogram bins or sample percentiles. The first three disjoint event counts satisfy $10 + 91 + 8 = 109$; the fourth event overlaps them.

For an event A with count k_A,

$$p_{\text{emp}} = \frac{k_A}{109}, \quad \text{EA} = |p_{\text{emp}} - p_{\text{model}}|, \quad E_{\%} = 100\frac{\text{EA}}{|p_{\text{model}}|}\%.$$

For the first event, $\text{EA} = |10/109 - 0.099999993325| = 0.008256874$ and $E_{\%} = 8.26\%$.

The difference is 0.826 percentage points, distinct from the relative percentage error. No model probability is zero.

This comparison implements C6 (relative frequencies versus theoretical probabilities); the equality of sample and fitted moments implements the estimation convention used in C7, not an independent model check.



Figure 4: Errors for the four events in Table 4. Teal bars and the left scale show absolute probability errors; violet bars and the right scale show relative percentage errors. Bar heights use unrounded values; labels show two decimal places.

6 Interpret the results

The fitted normal probability of height above 180 cm is 0.126620 (12.66%); the observed frequency is $13/109 = 0.119266$ (11.93%). The descriptive mixture gives 0.114421 (11.44%). Both are plausible in-sample approximations; the mixture is closer at this cutoff, which alone does not establish better prediction for new students.

For the normal model, the unique largest absolute difference occurs at $157.858453 \leq X \leq 181.269987$ cm: 0.034862 (3.49 percentage points). Imperfect symmetry or student-subgroup heterogeneity may contribute, but neither is established as the cause.

Table 5: Principal limitations

Issue	Implication
Sampling / measurement	Unknown selection and recording procedures limit population generalization.
Model support / shape	The normal density permits impossible negative or arbitrarily large heights; histogram symmetry is imperfect.
Operational applicability	The 180 cm cutoff is hypothetical; height alone does not determine uniform size.
In-sample evaluation	The same 109 records fit and assess the model; agreement does not establish predictive accuracy for new students.

7 Register AI use

Conversation 1
Conversation 2
Conversation 3
Conversation 4
Conversation 5

8 Brief extension: two-component skew-normal mixture

As a descriptive alternative, use the already fitted model in `assets/data/heights.tex`, with full-precision parameters and quantiles from `heights_results.json`; no model is refitted here. It combines two curves, with

weights adding to one:

$$f_M(x) = \sum_{j=1}^{2} w_j \frac{2}{\omega_j} \phi(z_j)\Phi(\alpha_j z_j), \qquad z_j = \frac{x - \xi_j}{\omega_j}.$$

Here ϕ and Φ are the standard-normal density and CDF; ξ is location, $\omega > 0$ is scale, and α controls skewness. Skew-normal location and scale are generally *not* the component mean and SD. For example, component 1 has mean 160.9511 cm and SD 4.7685 cm, despite its different location and scale below.

Table 6: Mixture parameters (rounded only for display)

Component	Weight w	Location ξ (cm)	Scale ω (cm)	Shape α
1	0.444318	166.734909	7.496063	-3.797348
2	0.555682	169.598283	8.223408	20.000000

Table 7: Sample and fitted-model moments

Quantity	Sample	Fitted normal	Fitted mixture
Mean (cm)	169.564220	169.564220	169.397656
Variance (cm^2)	83.431023	83.431023	80.863867
SD (cm)	9.134058	9.134058	8.992434

The sample variance uses $n - 1$; model variances are distribution moments. The normal column uses the main task's sample-SD rule. The source's model-selection table instead uses the maximum-likelihood normal SD ($\sqrt{\sum(x_i - \bar{x})^2/n}$). Among its candidate models, AIC favors two normals and BIC favors a single normal; neither selects this mixture.

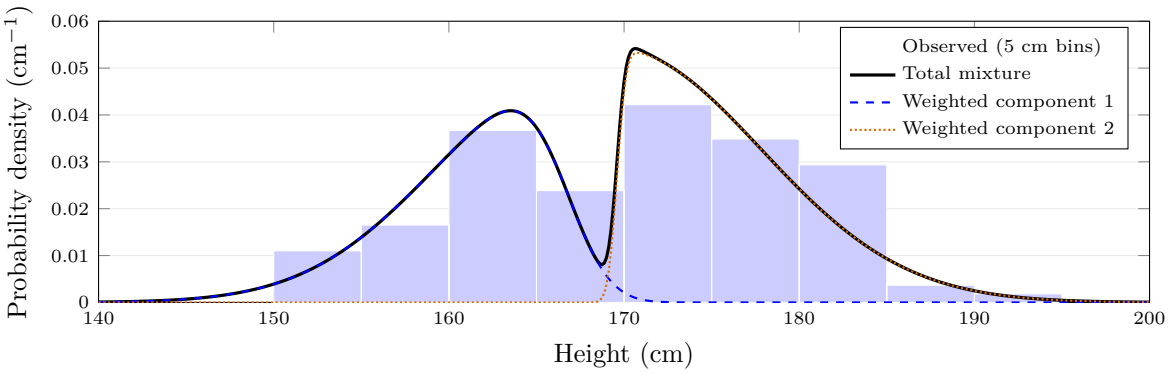


Figure 5: Fitted skew-normal mixture and its weighted components over the observed histogram. Bins are $[150, 155), \ldots, [190, 195)$; each bar height is its count divided by 109×5. Bars have total area one; curves and bars share density units. Both density overlays use identical axes and bins.

The mixture CDF is $F_M(t) = \sum_j w_j F_j(t)$, where F_j is the skew-normal CDF for component j. Its 10th and 90th percentiles are

$$a_M = q_{0.10} = 157.640900\,\text{cm}, \qquad b_M = q_{0.90} = 180.624899\,\text{cm}.$$

These displayed boundaries are rounded: all mixture counts and CDF evaluations below use the *same full-precision JSON boundaries* ($a_M = 157.6408995842728$, $b_M = 180.62489857403529$ cm). The normal checks in Table 4 retain their own numerical cutoffs $a = 157.858453$, $b = 181.269987$ cm. Thus the two central-interval checks concern different events; only $X > 180$ is a shared threshold. Both models use strict lower/upper tails and an inclusive central interval. Each empirical/model pair uses exactly matching endpoints; continuous models assign zero probability to individual endpoints.

Counts $10 + 86 + 13 = 109$ partition the observations; $X > 180$ overlaps them. The mixture's own quantile regions have fitted probabilities 0.10, 0.80, 0.10 by construction. Individual observations, rather than histogram bins, determine all counts.

Errors in model moments. For each moment T (mean, variance or SD), use the sample statistic as the reference and define

$$E_{A,T} = |T_{\text{model}} - T_{\text{sample}}|, \qquad E_{\%,T} = 100\frac{E_{A,T}}{|T_{\text{sample}}|}\%.$$

Table 8: Mixture checks on the same 109 individual observations; EA and relative error use the main task's definitions

Event (cm)	Count	Empirical	Mixture	EA	Error (%)
$X < a_M$	10	0.091743	0.100000	0.008257	8.26
$a_M \le X \le b_M$	86	0.788991	0.800000	0.011009	1.38
$X > b_M$	13	0.119266	0.100000	0.019266	19.27
$X > 180$	13	0.119266	0.114421	0.004845	4.23

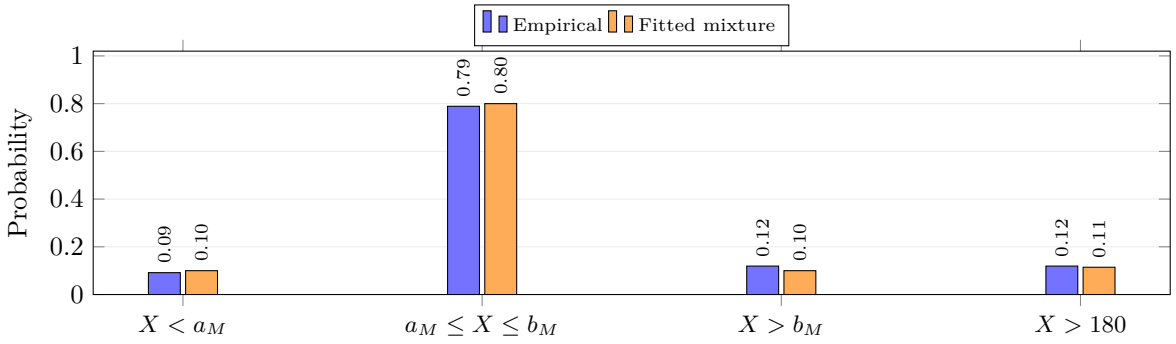


Figure 6: Paired mixture probabilities at matching endpoints from Table 8; labels show two decimals, as in Figure 3.

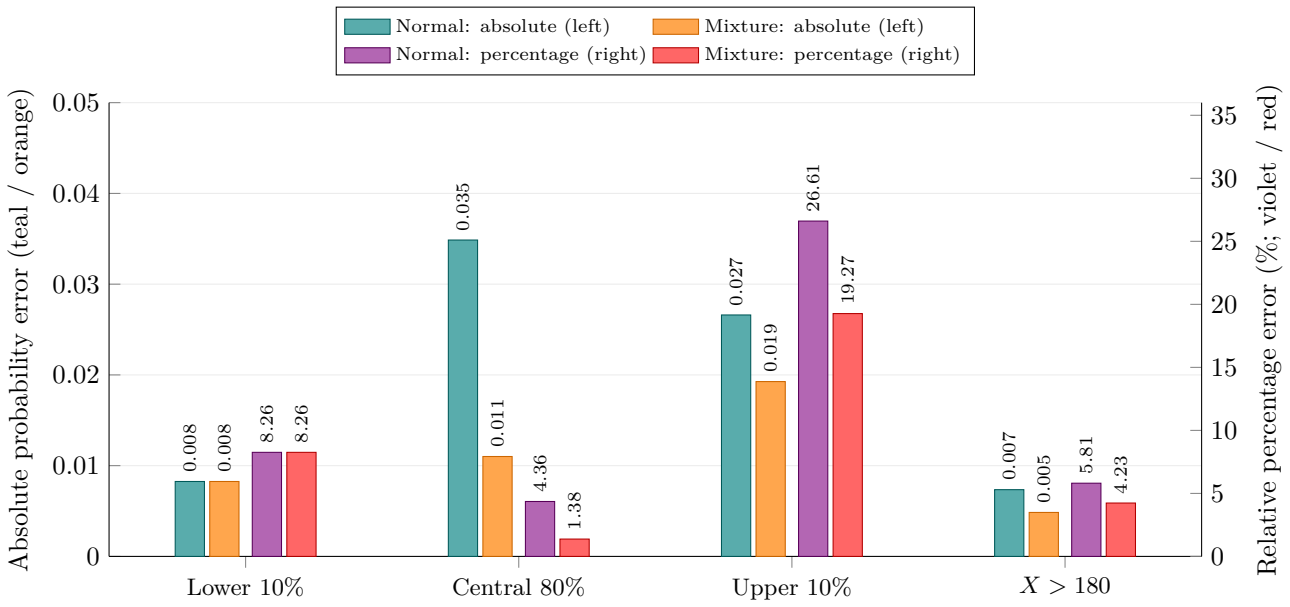


Figure 7: Errors for the normal fit (Table 4) and the skew-normal mixture (Table 8), grouped by event. Each four-bar group contains two close pairs: normal and mixture absolute errors (teal and orange, left scale), then normal and mixture percentage errors (violet and red, right scale). The normal colors match Figure 4. Heights across the two scales are not directly comparable. The first three groups use each model's own quantile boundaries (a, b for the normal; a_M, b_M for the mixture), so they concern different numerical events. Only $X > 180$ uses the same threshold in both models. Bar heights use unrounded errors; labels show three decimals for absolute errors and two for percentage errors.

These moment percentage errors use the sample reference; the probability percentage errors above use the model probability as their reference. Figure 8 compares the moments in Table 7 using full-precision values.

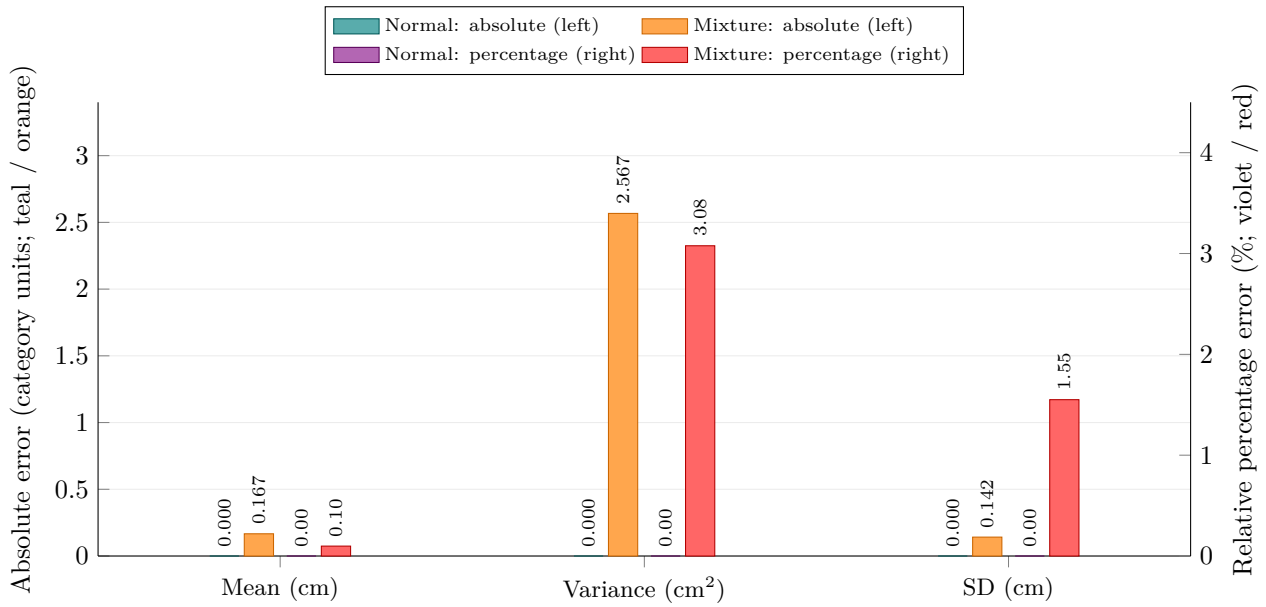


Figure 8: Errors in the model mean, variance and SD relative to the sample statistics. Each group contains two close pairs: normal and mixture absolute errors (teal and orange, left scale), then normal and mixture percentage errors (violet and red, right scale), matching Figure 7. The normal errors are zero by construction because its parameters use the sample mean and SD; this is not independent evidence of fit. Absolute errors have the units shown for each category (cm for mean and SD; cm^2 for variance), so their heights across categories measure different quantities. The right scale shows percentage errors on a common relative scale; heights across the left and right scales are not directly comparable. Labels show three decimals for absolute errors and two for percentage errors.

The mixture's central-region absolute difference is 0.011009, compared with the normal's 0.034862 at its different boundaries. This does not compare identical events. At the shared 180 cm threshold, the mixture gives $P_M(X > 180) = 0.114421$, compared with 0.126620 for the normal and $13/109 = 0.119266$ empirically. Both estimates are close to the observed proportion; the mixture has the smaller absolute difference at this threshold.

The mixture can describe asymmetry and two peaks, but the second shape parameter reaches the fitted bound ($\alpha_2 = 20$), so its sharp features depend on a modeling constraint. Its components do not establish demographic groups. Rounded heights, unknown sampling and measurement procedures, and using the same data for fitting and checking limit interpretation. In-sample agreement does not establish predictive superiority or make the mixture the preferred model.